



Lecture 06: Network Layer (I)

Network layer basics

IP protocol

Addressing



Application

Transport l.

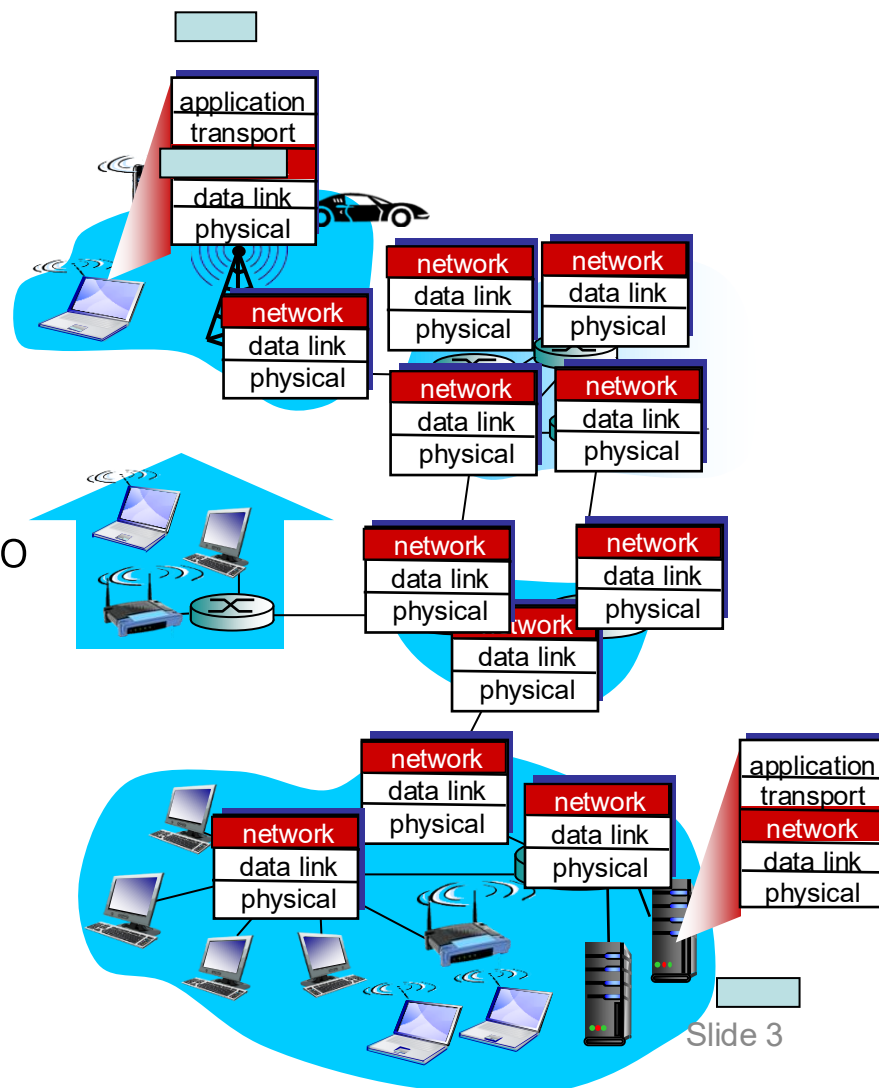
Network l.

Link l.

Functions of the network layer

Network layer

- transport segment from sending to receiving host
- on sending side encapsulates segments into datagrams
- on receiving side, delivers segments to transport layer
- network layer protocols in every host, router
- router examines header fields in all IP datagrams passing through it



Network-layer functions

- **forwarding**: move packets from router's input to appropriate router output *data plane*
- **routing**: determine route taken by packets from source to destination *control plane*

Two approaches to structuring network control plane:

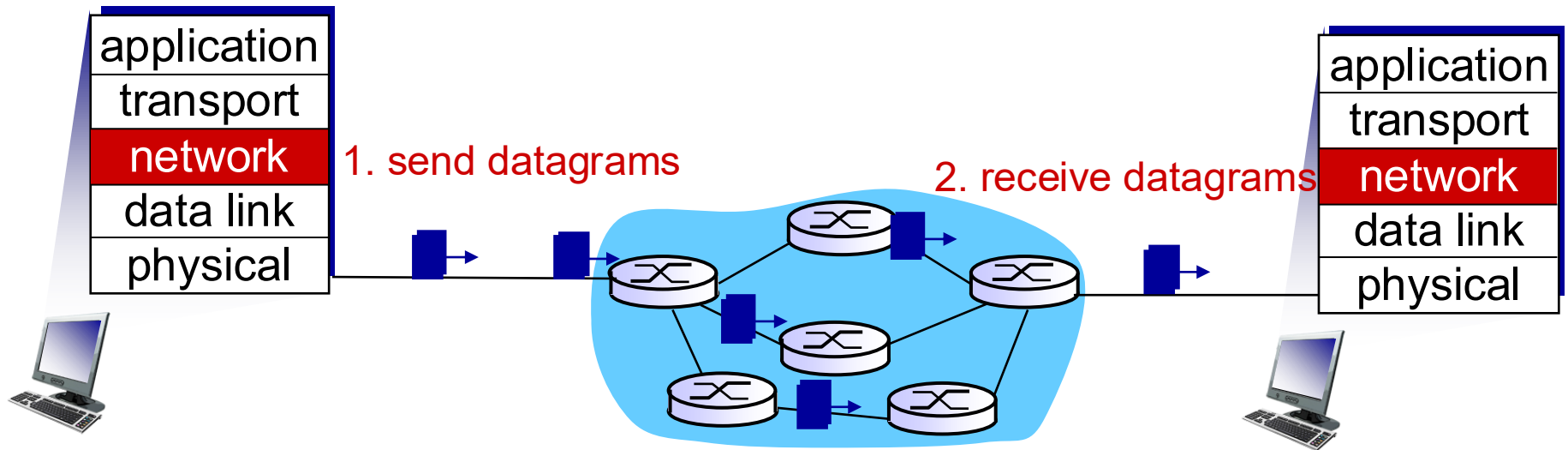
- per-router control (traditional)
- logically centralized control (software defined networking)



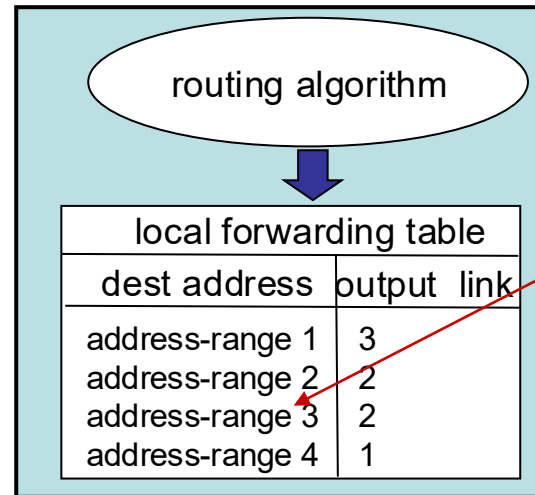
Datagram networks

Datagram networks

- no call setup at network layer
- routers: no state about end-to-end connections
 - no network-level concept of “connection”
- packets forwarded using destination host address

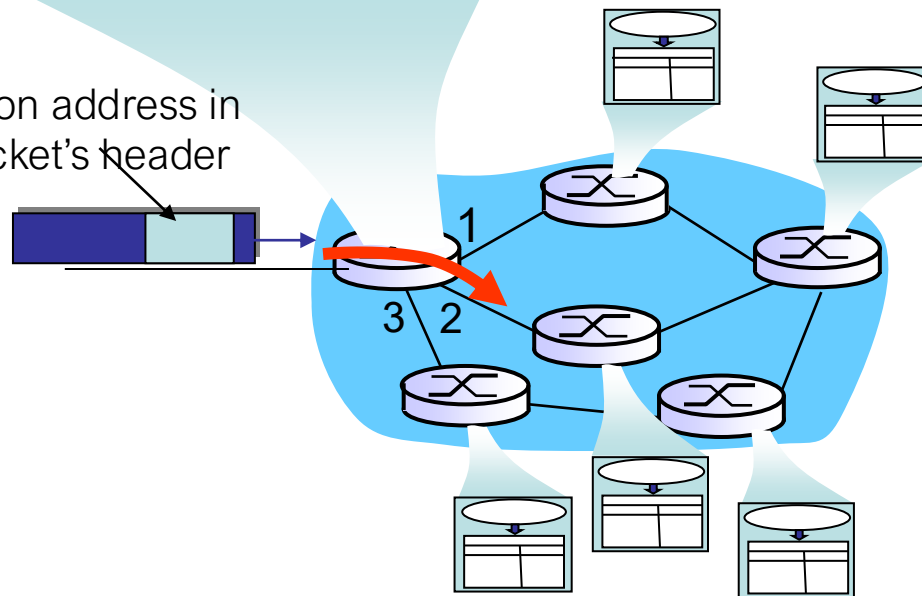


Forwarding table



6 billion IP addresses, so rather than list individual destination address
list range of addresses (aggregate table entries)

IP destination address in arriving packet's header



Datagram forwarding table

Destination Address Range	Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011001 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

Longest prefix matching

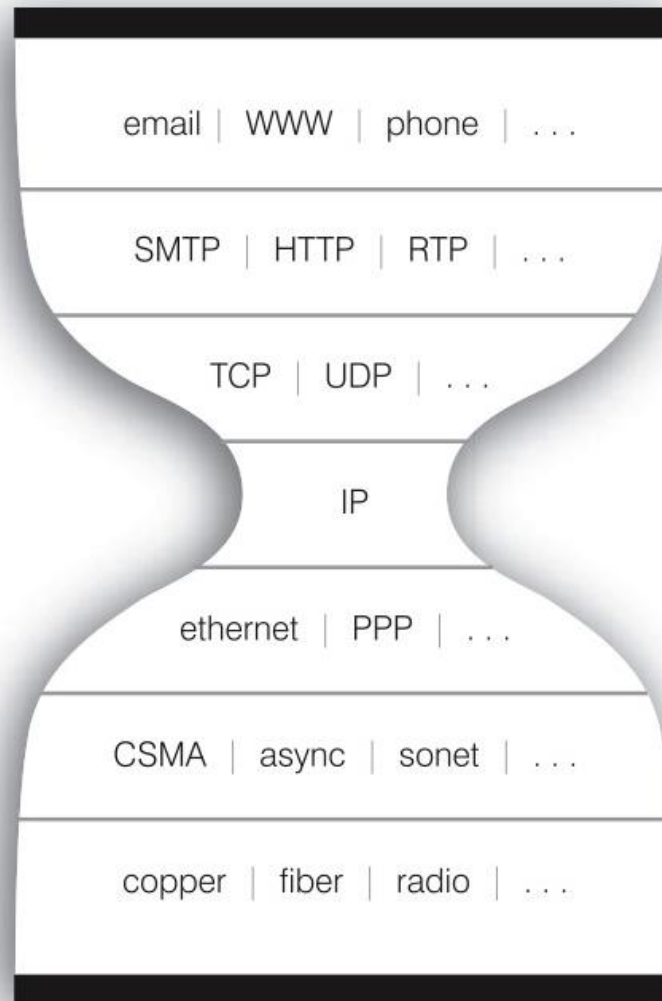
when looking for forwarding table entry for given destination address, use **longest** address prefix that matches destination address.

Destination Address Range	Link interface
11001000 00010111 00010*** *****	0
11001000 00010111 00011000 *****	1
11001000 00010111 00011*** *****	2
otherwise	3



IP protocol

Narrow waist of the Internet



IP functionalities

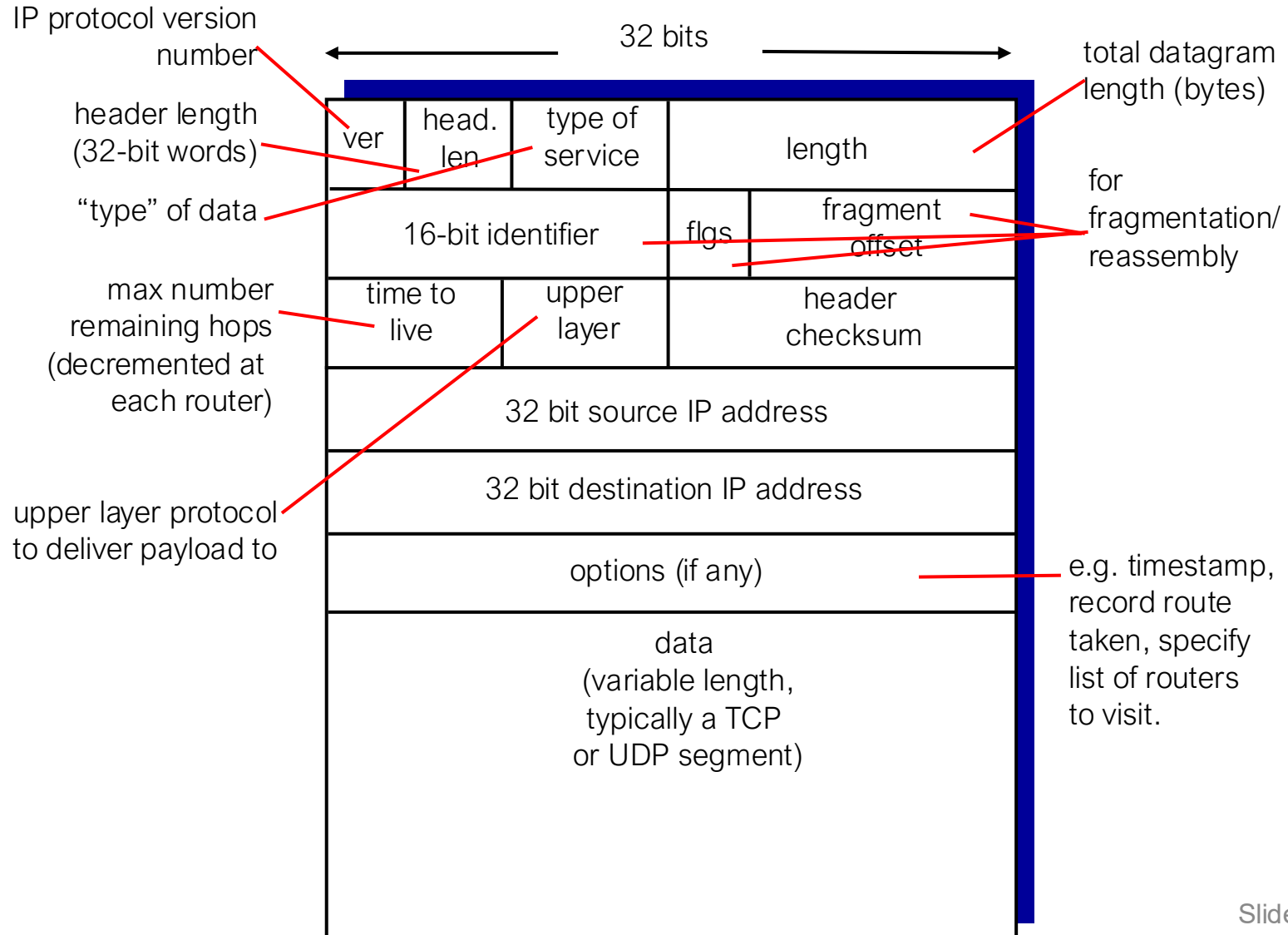
Four main characteristics:

- A connectionless service
- Unreliable delivery
- A best-effort delivery
- No in-sequence delivery



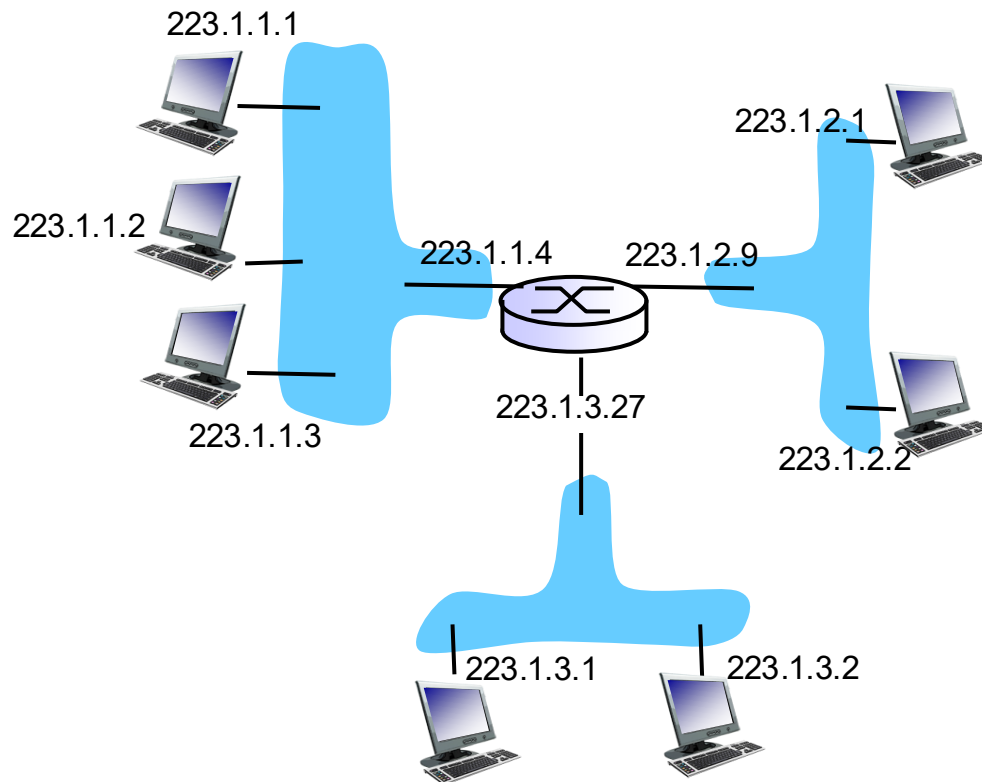
IPv4

IPv4 datagram format



IPv4 address

- **IP address**: 32-bit identifier for host/router interface
- **interface**: connection between host/router and physical link
 - router's typically have multiple interfaces
 - host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)
- 1 IP address associated with each interface



223.1.1.1 = 11011111 00000001 00000001 00000001

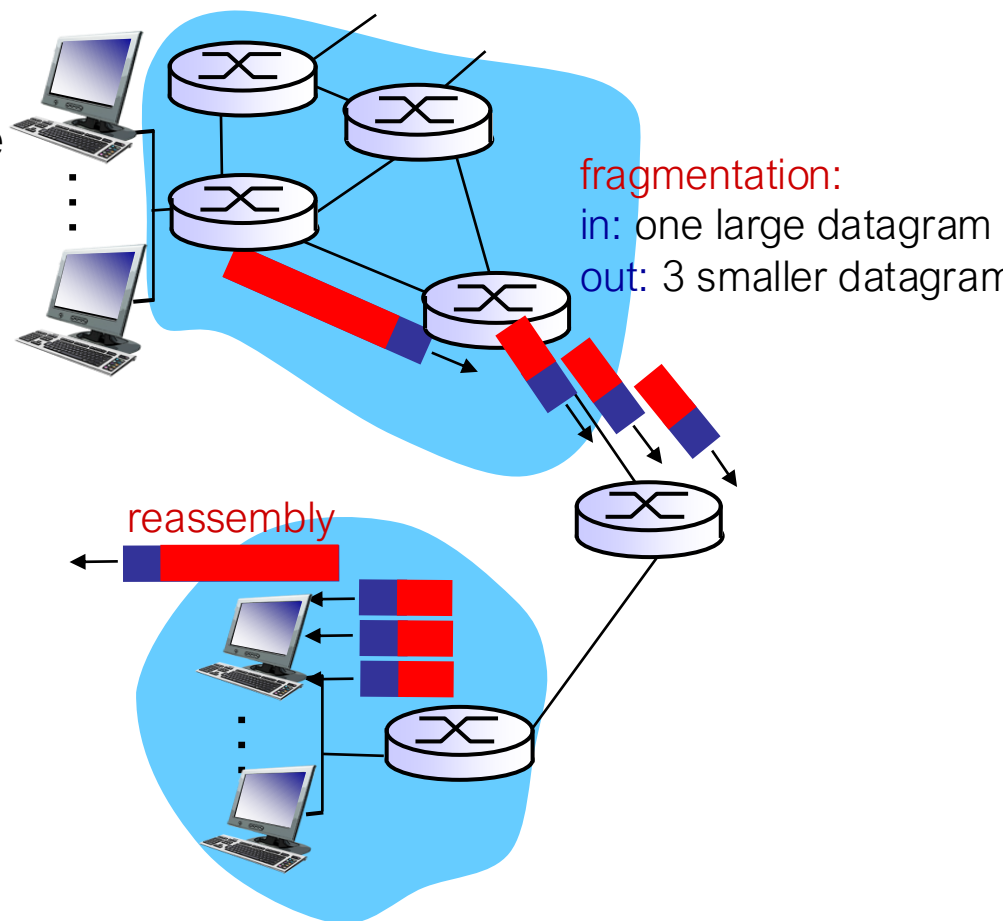
223
1
1
1



Fragmentation

IP fragmentation and reassembly

- network links have MTU (maximum transmission unit) - largest possible link-level frame
 - different link types, different MTUs
- large IP datagram divided (“fragmented”) within net
 - one datagram becomes several datagrams
 - “reassembled” only at final destination



Example

- ❖ 4000 byte datagram
- ❖ MTU = 1500 bytes

	length	ID	fragflag	offset
	=4000	=x	=0	=0

one large datagram becomes
several smaller datagrams

1480 bytes in
data field

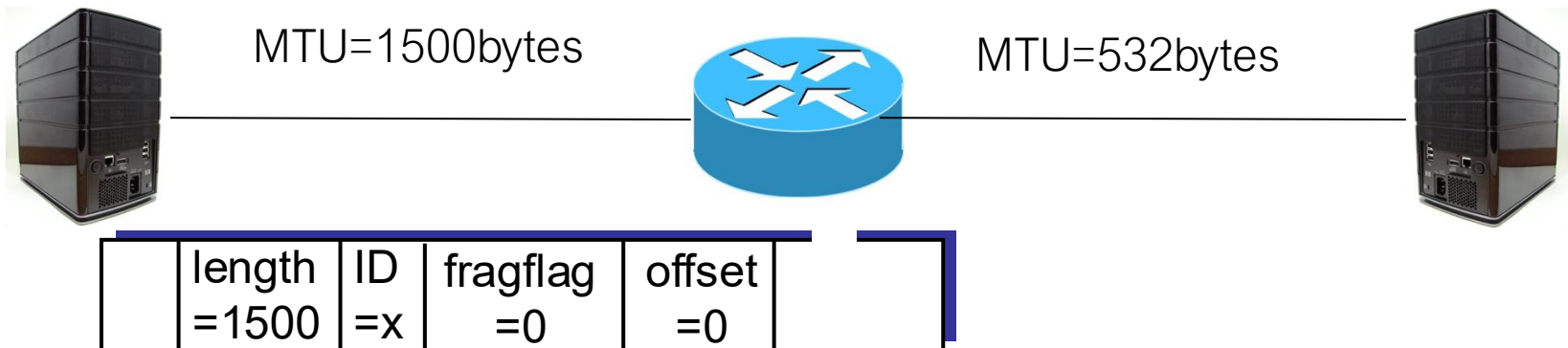
offset =
 $1480/8$

	length	ID	fragflag	offset
	=1500	=x	=1	=0

	length	ID	fragflag	offset
	=1500	=x	=1	=185

	length	ID	fragflag	offset
	=1040	=x	=0	=370

Quiz



How will the fragments look like?

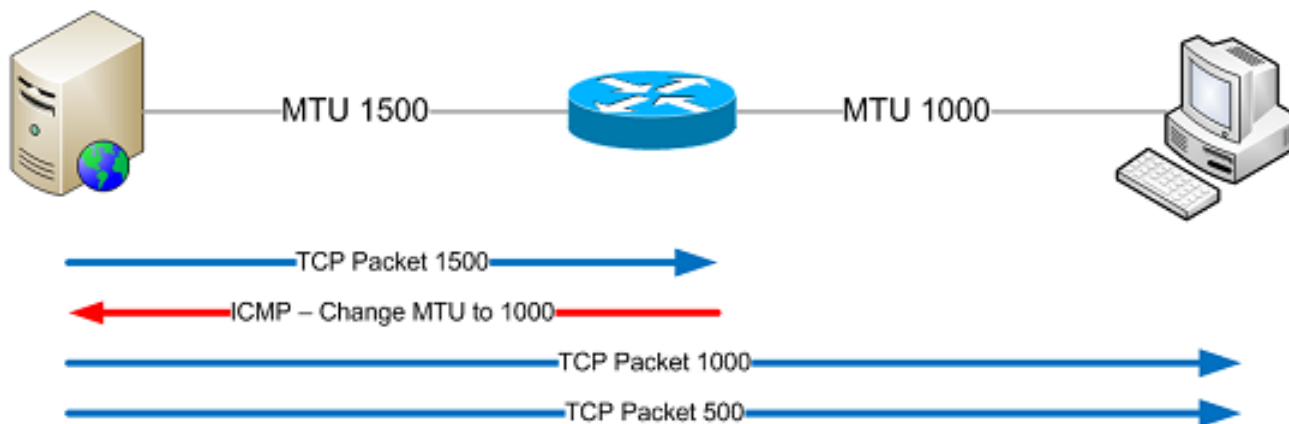
	length =?	ID =x	fragflag =?	offset =?	
	length =?	ID =x	fragflag =?	offset =?	

Avoid it if you can

- Fragments may be lost
- Fragments reduce throughput.
- Fragments need to be reassembled: this can costly.

Avoiding fragmentation:

- Path MTU Discovery (PMTUD)





DHCP

Getting an IP address

1. Hard-coded by system admin in a file
 - Windows: control-panel->network->configuration->tcp/ip->properties
 - UNIX: /etc/rc.config

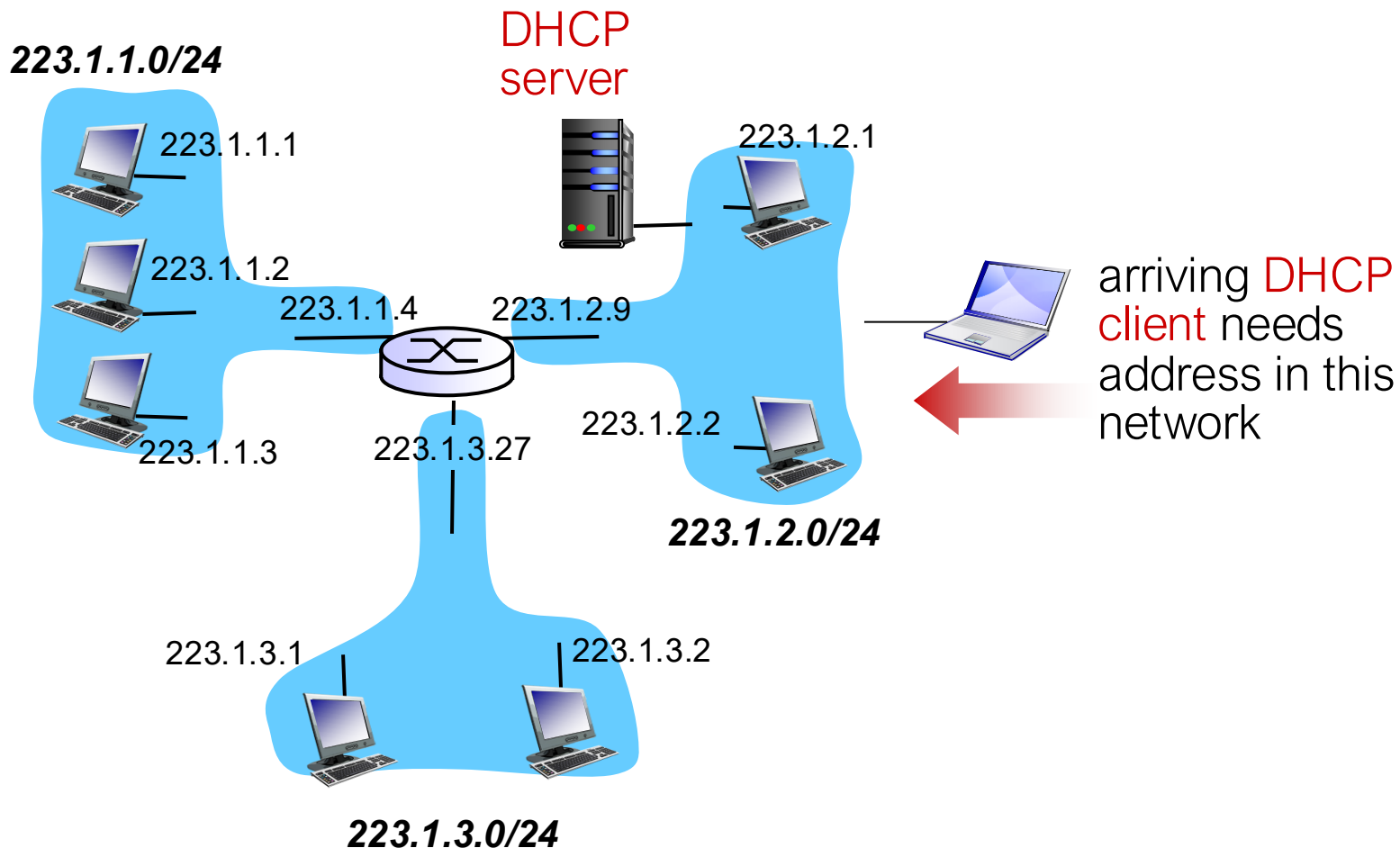
2. **DHCP**: **D**ynamic **H**ost **C**onfiguration **P**rotocol: dynamically get address from as server
 - “plug-and-play”

DHCP

Dynamic Host Configuration Protocol

- It allows host to dynamically obtain its IP address from network server when it joins network
 - can renew its lease on address in use
 - allows reuse of addresses (only hold address while connected/“on”)
 - support for mobile users who want to join network (more shortly)
- DHCP overview:
 - host broadcasts “DHCP discover” msg [optional]
 - DHCP server responds with “DHCP offer” msg [optional]
 - host requests IP address: “DHCP request” msg
 - DHCP server sends address: “DHCP ack” msg

DHCP client-server scenario



DHCP client-server scenario

DHCP server: 223.1.2.5

DHCP discover

src : 0.0.0.0, 68
dest.: 255.255.255.255,67
yiaddr: 0.0.0.0
transaction ID: 654

arriving
client



DHCP offer

src: 223.1.2.5, 67
dest: 255.255.255.255, 68
yiaddr: 223.1.2.4
transaction ID: 654
lifetime: 3600 secs

DHCP request

src: 0.0.0.0, 68
dest.: 255.255.255.255, 67
yiaddr: 223.1.2.4
transaction ID: 655
lifetime: 3600 secs

DHCP ACK

src: 223.1.2.5, 67
dest: 255.255.255.255, 68
yiaddr: 223.1.2.4
transaction ID: 655
lifetime: 3600 secs



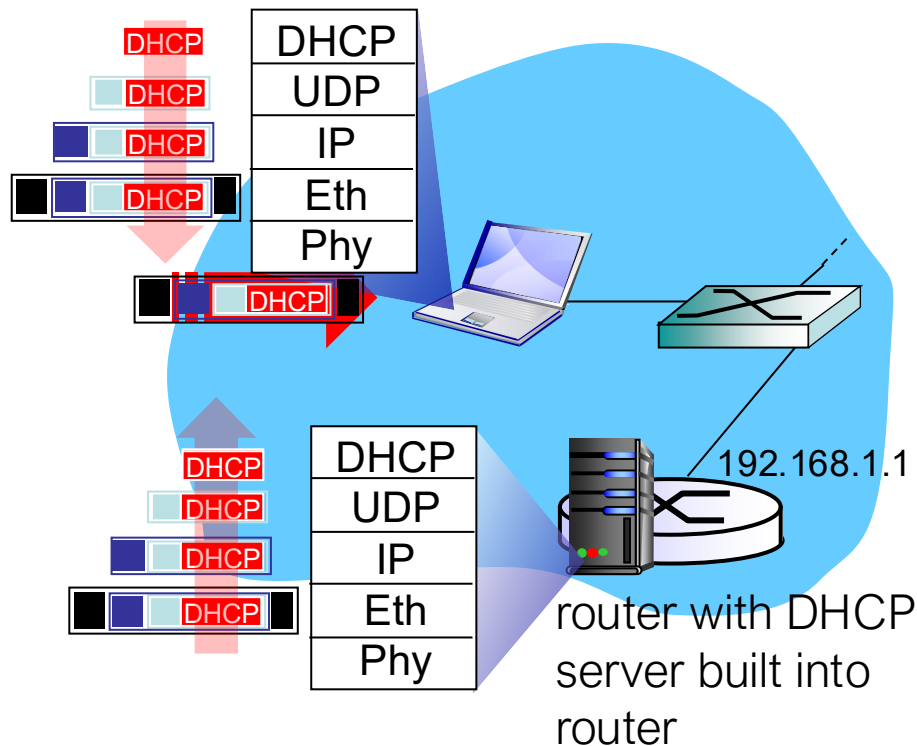
DHCP option

DHCP can return more than just allocated IP address on subnet.
For this DHCP uses the option fields.

- Address of first-hop router for client
- Network mask (indicating network versus host portion of address)
- Name and IP address of DNS server
- Name and IP address of the time servers

DHCP: example

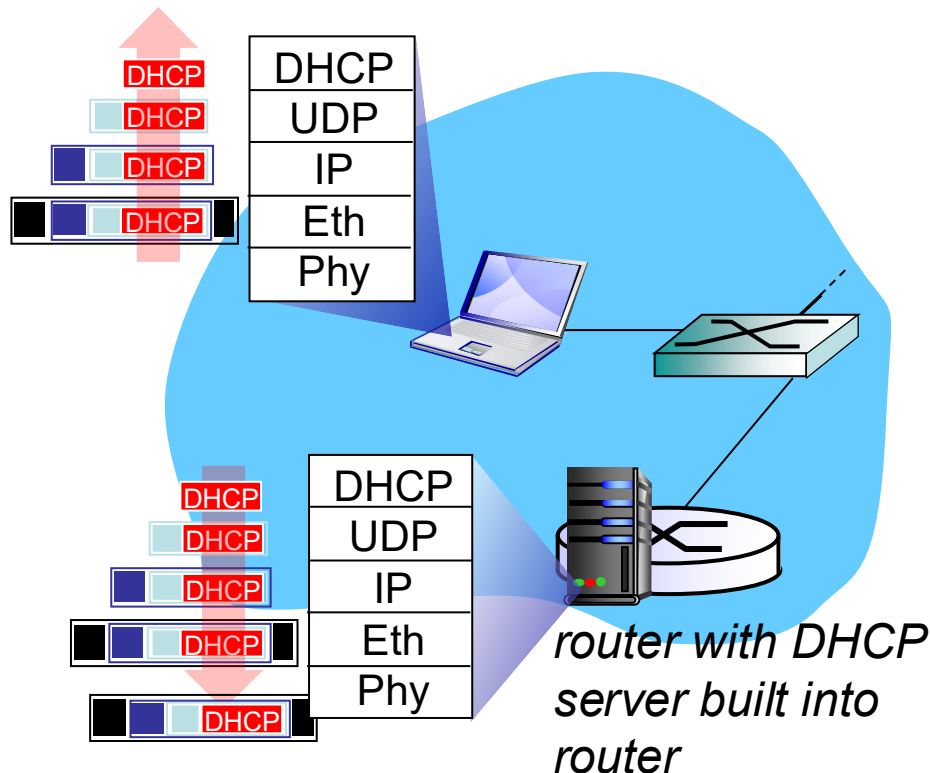
- connecting laptop needs its IP address, addr of first-hop router, addr of DNS server: use DHCP



- ❖ DHCP request encapsulated in UDP, encapsulated in IP, encapsulated in 802.1 Ethernet
- ❖ Ethernet frame broadcast (dest: FFFFFFFFFFFFFFFF) on LAN, received at router running DHCP server
- ❖ Ethernet demuxed to IP demuxed, UDP demuxed to DHCP

DHCP: example

- DHCP server formulates DHCP ACK containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server



- ❖ encapsulation of DHCP server, frame forwarded to client, demuxing up to DHCP at client
- ❖ client now knows its IP address, name and IP address of DNS server, IP address of its first-hop router

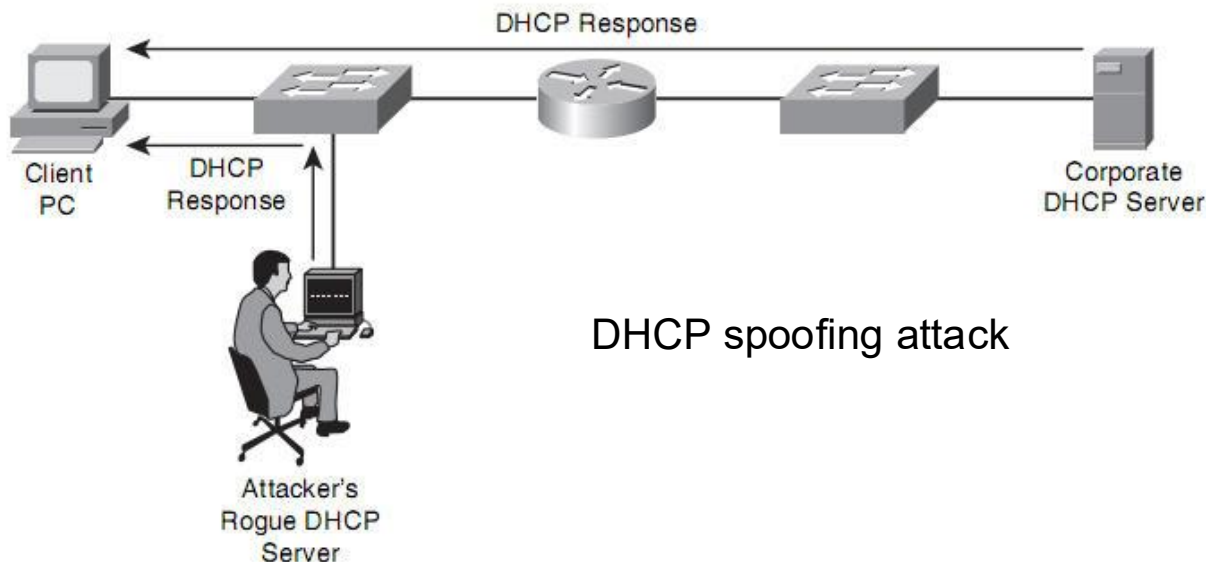
DHCP snooping

Need to define **trusted** and **untrusted** sources.

It acts like a firewall between untrusted nodes and the DHCP server.

Trusted ports can send requests and acks.

Untrusted ports can only forward DHCP requests.





Quiz Time

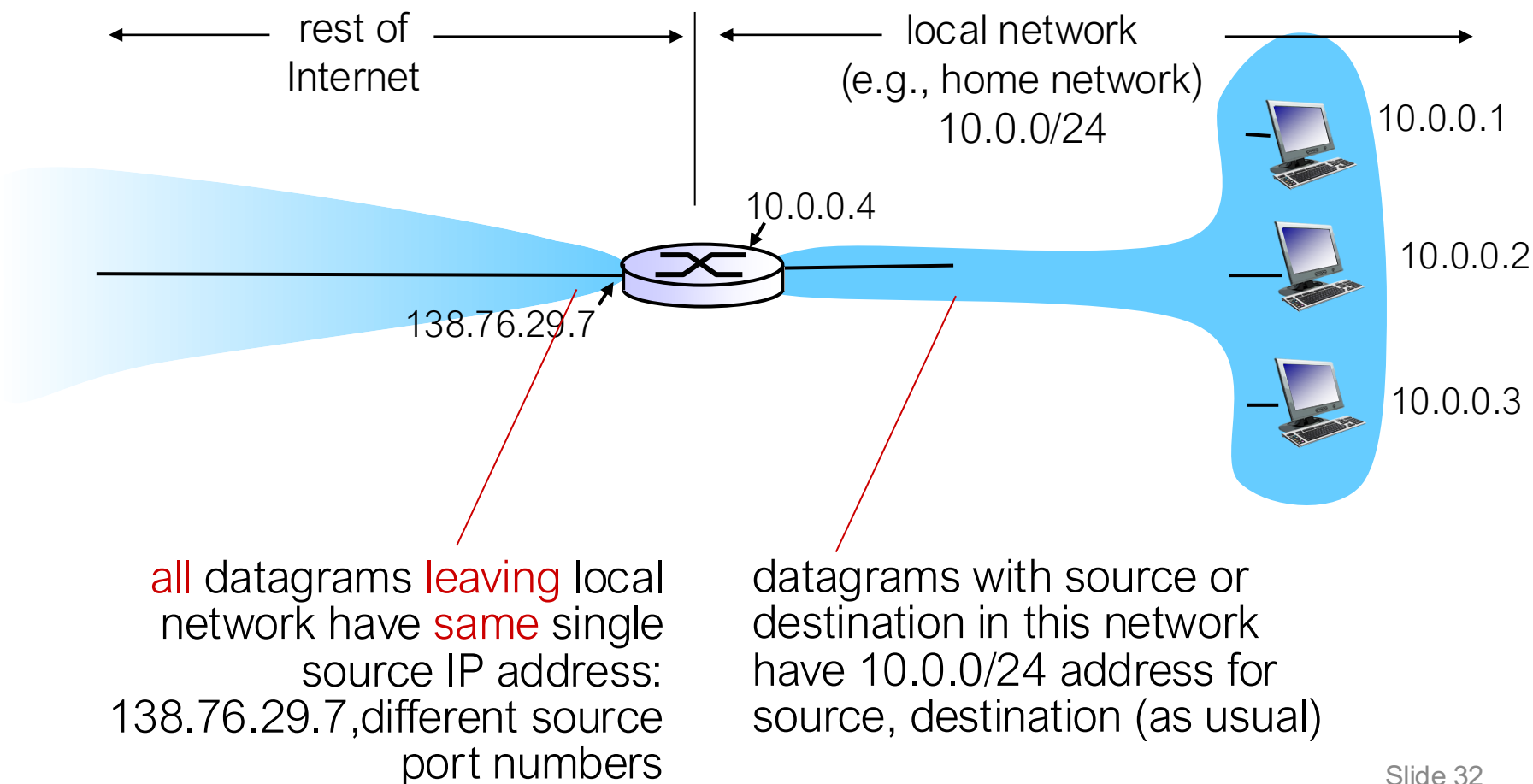
Let's begin with a (**graded**) Canvas quiz.
You have 5 minutes to complete the quiz. Start now.

Any questions afterwards: TicketVise → label “Review quizzes”



NAT

NAT: network address translation



NAT motivation

Local network uses just one IP address as far as outside world is concerned:

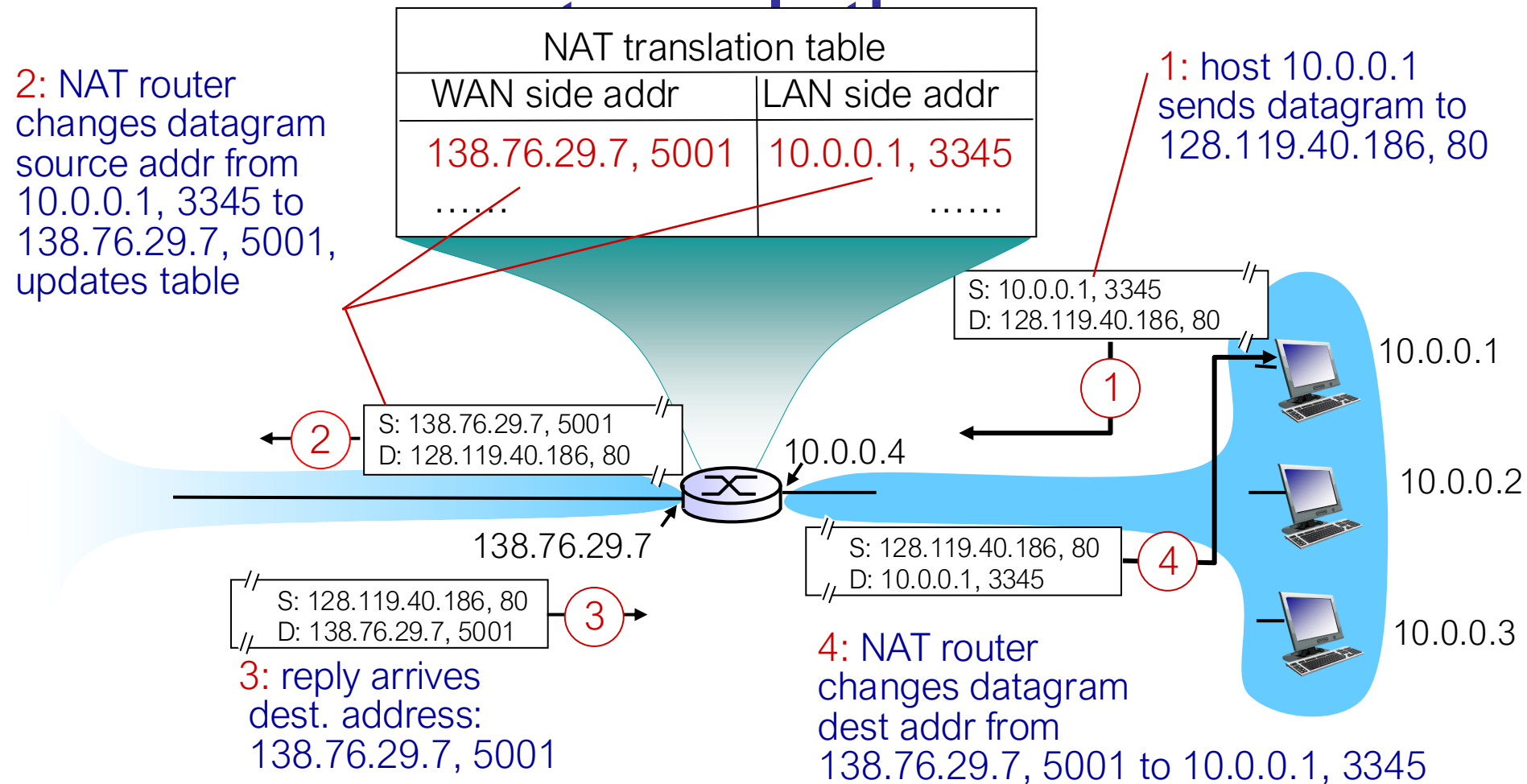
- range of addresses not needed from ISP: just one IP address for all devices
- can change addresses of devices in local network without notifying outside world
- can change ISP without changing addresses of devices in local network
- devices inside local net not explicitly addressable, visible by outside world (a security plus)

NAT: implementation

A NAT router must:

- **outgoing datagrams**: replace (source IP address, port #) of every outgoing datagram to (NAT IP address, new port #)
 - . . . remote clients/servers will respond using (NAT IP address, new port #) as destination addr
- remember (**in NAT translation table**) every (source IP address, port #) to (NAT IP address, new port #) translation pair
- **incoming datagrams**: replace (NAT IP address, new port #) in dest fields of every incoming datagram with corresponding (source IP address, port #) stored in NAT table

NAT: network address

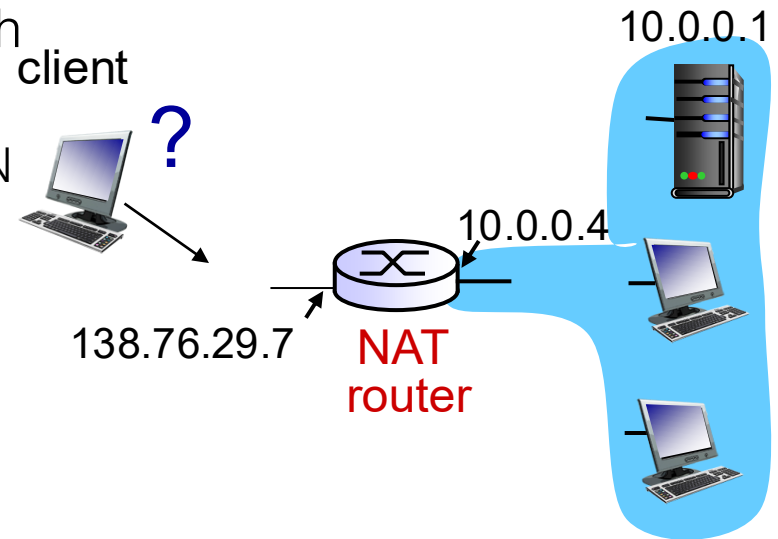


NAT controversy

- NAT is controversial:
 - routers should only process up to layer 3
 - violates end-to-end argument
 - NAT possibility must be taken into account by app designers, e.g., P2P applications
 - address shortage should instead be solved by IPv6

NAT traversal problem

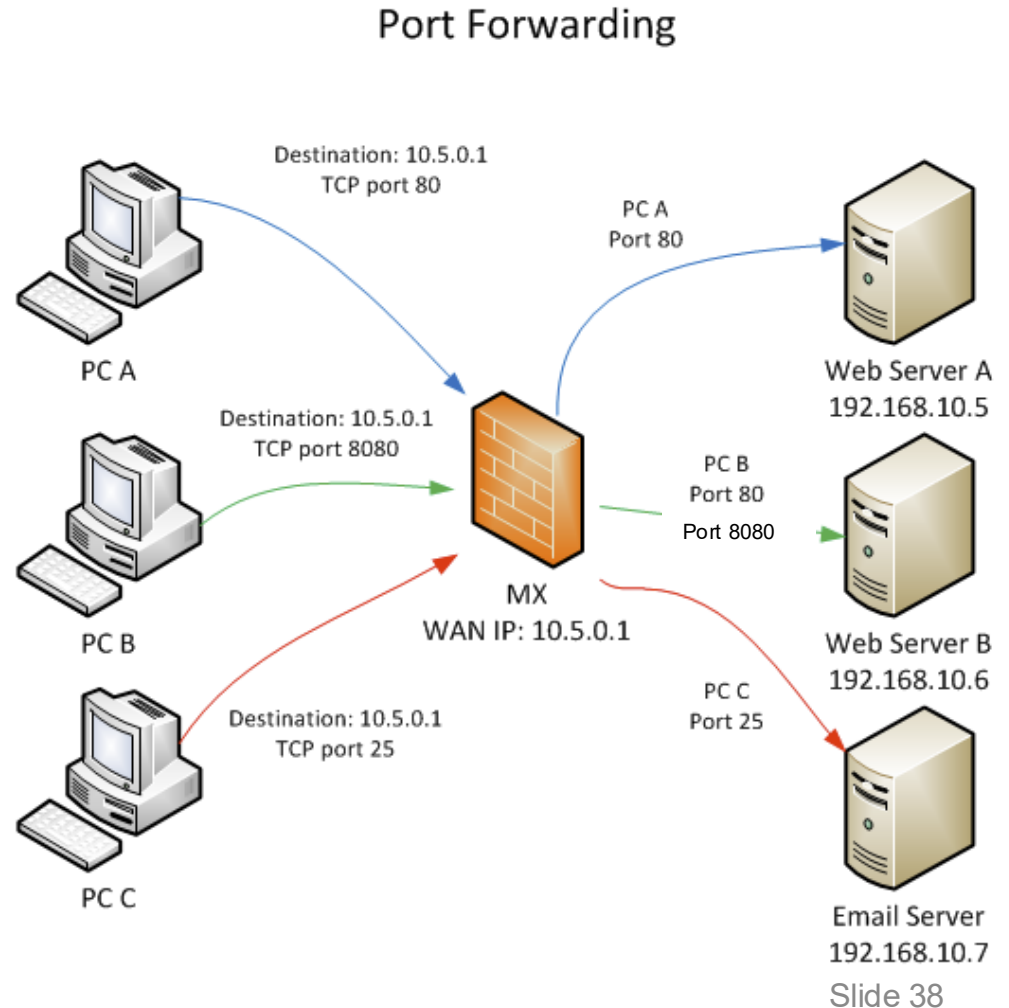
- client wants to connect to server with address 10.0.0.1
 - server address 10.0.0.1 local to LAN (client can't use it as destination addr)
 - only one externally visible NATed address: 138.76.29.7



Solution1: port forwarding

Configuring the gateway to send all packets received on a particular port to a specific machine on the internal network.

- e.g., (123.76.29.7, port 2500) always forwarded to 10.0.0.1 port 2500



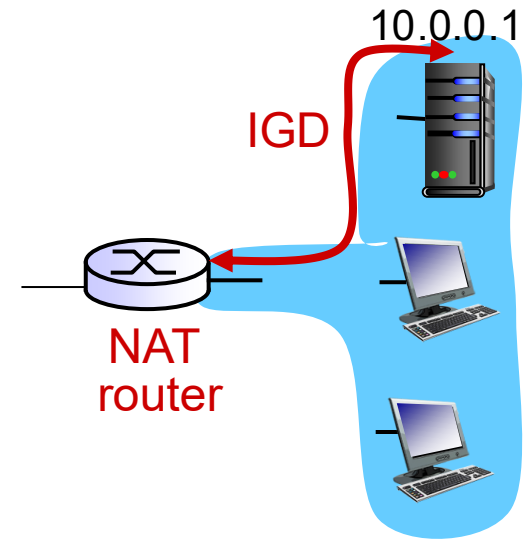
Solution 2: IGD

Part of the Universal Plug and Play (UPnP) family.

Internet Gateway Device (IGD) Protocol.

Allows NATed host to:

- learn public IP address (138.76.29.7)
- add/remove port mappings (with lease times)
- i.e., automate static NAT port map configuration



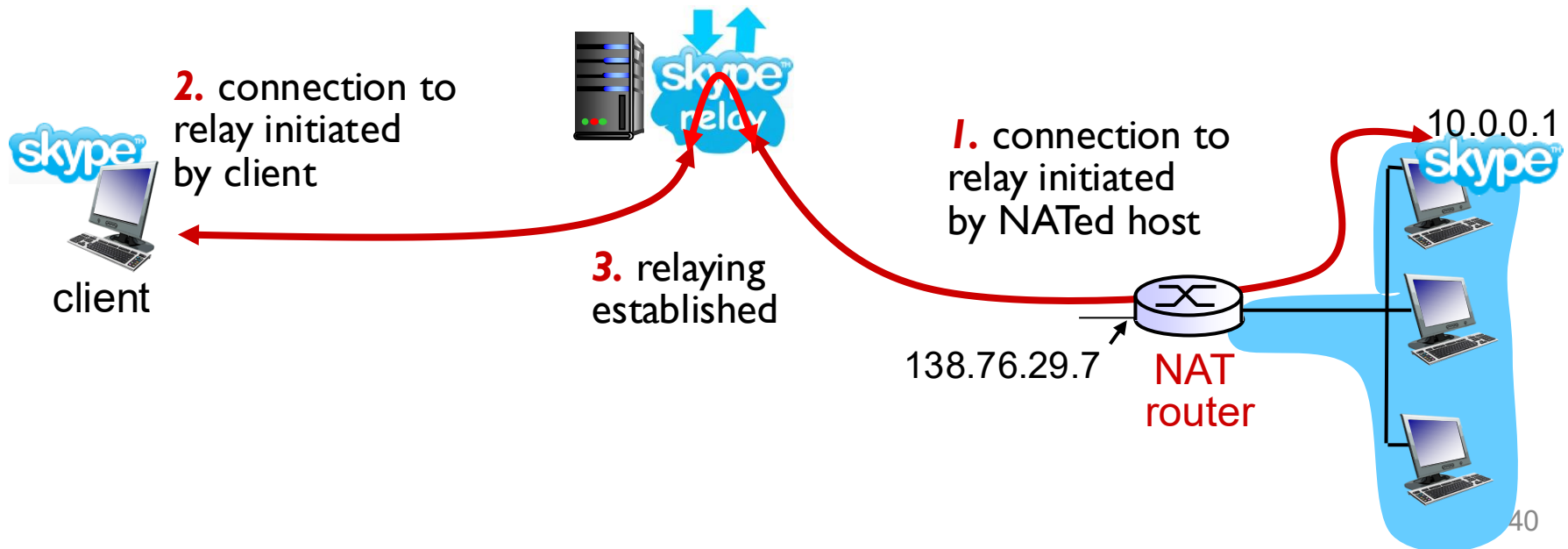
Solution 3: Relaying

It uses a relay

NATed client establishes connection to relay

external client connects to relay

relay bridges packets between to connections





IPv6 protocol

Motivation

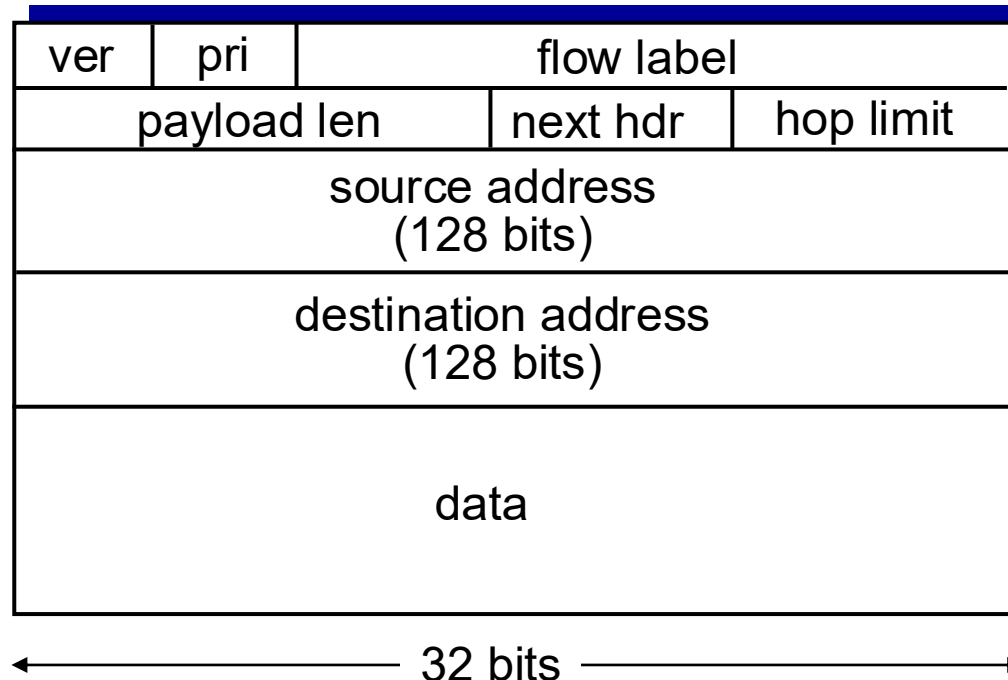
- **initial motivation:** 32-bit address space soon to be completely allocated.
- additional motivation:
 - header format helps speed processing/forwarding
 - header changes to facilitate QoS

IPv6 datagram format:

- fixed-length 40 byte header
- no fragmentation allowed

IPv6 datagram format

next header: identify upper layer protocol for data



Other changes from IPv4

- No fragmentation allowed
- **checksum**: removed entirely to reduce processing time at each hop
- **options**: allowed, but outside of header, indicated by “Next Header” field



IPv6 addressing

IPv6 addresses

Addresses are expressed as eight 16-bit values with colons between them, and each 16-bit value is displayed in hexadecimal.

For example: 2001:db8:31:1:20a:95ff:fe5:246e

Zeros are contracted (only once in an address!).

2001:db8:31:0:0:0:0:1 becomes 2001:db8:31::1

The loopback address 0:0:0:0:0:0:0:1 becomes ::1

Know more:

IP Version 6 Addressing Architecture

RFC 3513 - Apr. 2003

Basic addresses

An interface can have multiple addresses.

The basic types are:

- **Unicast** A single computer. A datagram will be delivered along the shortest path to this destination.

There are several forms of unicast address: global unicast address, the site-local address, the link-local address (fe80::), and the IPv4-capable host address.

- **Anycast** A set of computers (possibly at different destinations) sharing the same address. A datagram will be delivered along the shortest path to one member of the group.
- **Multicast** A set of computers (possibly at different destinations). One copy of the datagram will be delivered to all members of the group.

Note: in IPv6 broadcast is a special form of multicast.

Multicast addresses start with ff.

```
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=6463<RXCSUM,TXCSUM,TS04,TS06,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
    ether a0:78:17:7c:29:39
    inet6 fe80::18f6:ed09:7914:ce88%en0 prefixlen 64 secured scopeid 0xc
    inet 192.168.1.170 netmask 0xffffffff broadcast 192.168.1.255
    nd6 options=201<PERFORMNUD,DAD>
    media: autoselect
```

Stateless configuration

Uses link-local addresses and the embedded interface identifiers feature.
A host coming on the network:

1. Generates a link-local address `fe80::<interface_ID>`
2. Verifies its uniqueness
3. Solicits further information from the router with router solicitations
4. A routers responds with router advertisements, with prefixes that can be used to form global addresses

Example

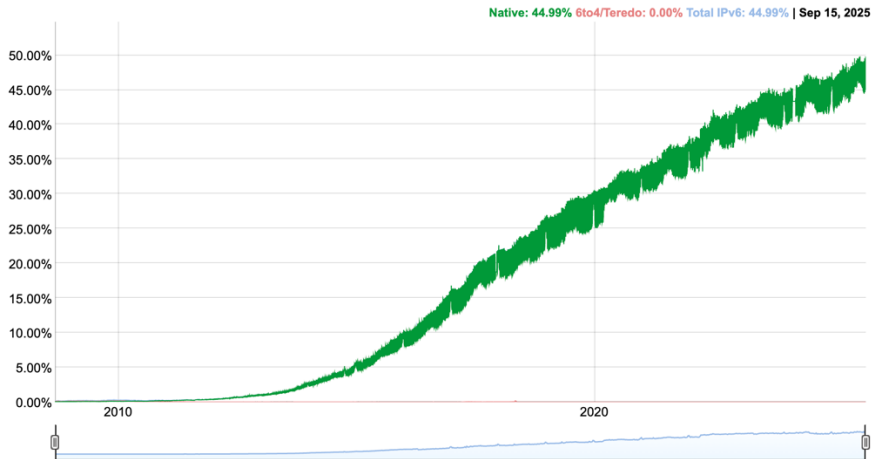
Prefix: `2001:db8:abcd:1::/64`

IP: `2001:db8:abcd:1:20a:95ff:fef5:246e`

IPv6: adoption

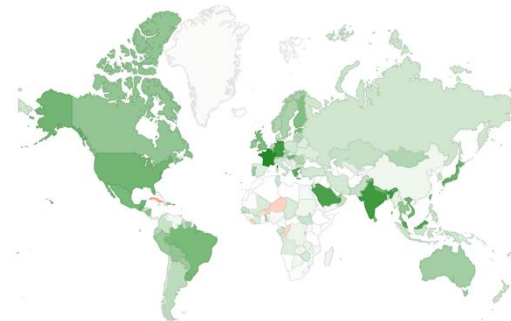
IPv6 Adoption

We are continuously measuring the availability of IPv6 connectivity among Google users. The graph shows the percentage of users that access Google over IPv6.



¹ <https://www.google.com/intl/en/ipv6/statistics.html>

Per-Country IPv6 adoption



[World](#) | [Africa](#) | [Asia](#) | [Europe](#) | [Oceania](#) | [North America](#) | [Central America](#) | [Caribbean](#) | [South America](#)

The chart above shows the availability of IPv6 connectivity around the world.

- Regions where IPv6 is more widely deployed (the darker the green, the greater the deployment) and users experience infrequent issues connecting to IPv6-enabled websites.
- Regions where IPv6 is more widely deployed but users still experience significant reliability or latency issues connecting to IPv6-enabled websites.
- Regions where IPv6 is not widely deployed and users experience significant reliability or latency issues connecting to IPv6-enabled websites.

IPv6: adoption

- Google¹: ~ 50% of clients access services via IPv6
- NIST: 1/3 of all US government domains are IPv6 capable
- Long (long!) time for deployment, use
 - 25 years and counting!
 - think of application-level changes in last 25 years: WWW, social media, streaming media, gaming, telepresence, ...
 - Why?

¹ <https://www.google.com/intl/en/ipv6/statistics.html>



ICMP

ICMP:

Internet Control Message Protocol

- Used by hosts & routers to communicate network-level information [RFC 792]
 - error reporting: unreachable host, network, port, protocol
 - echo request/reply (used by ping)
- Network-layer “above” IP:
 - ICMP msgs carried in IP datagrams
- ICMPv4 message: type, code, checksum plus data (IP header and first 8 bytes of IP datagram causing error)
- ICMPv6 message: type, code, checksum plus data

ICMPv4 types/codes

<u>Type</u>	<u>Code</u>	<u>description</u>
0	0	echo reply (ping)
3	0	dest. network unreachable
3	1	dest host unreachable
3	2	dest protocol unreachable
3	3	dest port unreachable
3	6	dest network unknown
3	7	dest host unknown
4	0	source quench (congestion control - not used)
8	0	echo request (ping)
9	0	route advertisement
10	0	router discovery
11	0	TTL expired
12	0	bad IP header

ICMPv6 types/codes (I)

Type	Name	Reference
0	Reserved	
1	Destination Unreachable	[RFC4443]
2	Packet Too Big	[RFC4443]
3	Time Exceeded	[RFC4443]
4	Parameter Problem	[RFC4443]
100	Private experimentation	[RFC4443]
101	Private experimentation	[RFC4443]
102-126	Unassigned	
127	Reserved for expansion of ICMPv6 error messages	[RFC4443]
128	Echo Request	[RFC4443]
129	Echo Reply	[RFC4443]
130	Multicast Listener Query	[RFC2710]
131	Multicast Listener Report	[RFC2710]
132	Multicast Listener Done	[RFC2710]

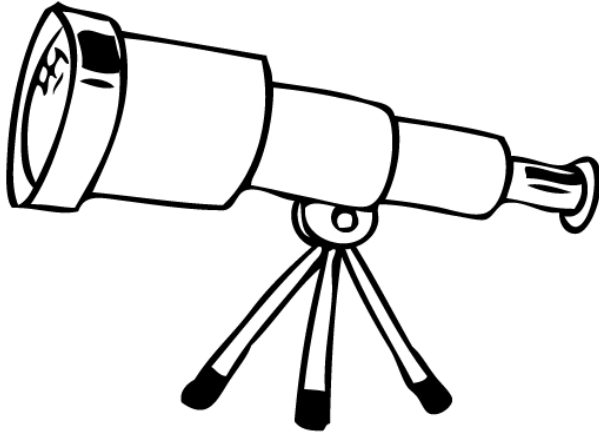
ICMPv6 types/codes (II)

Type	Name	Reference
133	Router Solicitation	[RFC4861]
134	Router Advertisement	[RFC4861]
135	Neighbor Solicitation	[RFC4861]
136	Neighbor Advertisement	[RFC4861]
137	Redirect Message	[RFC4861]
138	Router Renumbering	[Matt_Crawford]
139	ICMP Node Information Query	[RFC4620]
140	ICMP Node Information Response	[RFC4620]
141	Inverse Neighbor Discovery Solicitation Message	[RFC3122]
142	Inverse Neighbor Discovery Advertisement Message	[RFC3122]
143	Version 2 Multicast Listener Report	[RFC3810]
144	Home Agent Address Discovery Request Message	[RFC6275]
145	Home Agent Address Discovery Reply Message	[RFC6275]
146	Mobile Prefix Solicitation	[RFC6275]
147	Mobile Prefix Advertisement	[RFC6275]
148	Certification Path Solicitation Message	[RFC3971]



What have we learned today?

- Basic features of the network layer
- IPv4 and IPv6
- DHCP and NAT
- ICMP



What will we learn next time?

How do routing algorithms work?

Which routing algorithms are used in a network domain?

Material covered today

- Chapter 4:
 - Section 4.1 – Overview of the network layer
 - Section 4.3 – The Internet Protocol (IP)
- Chapter 5:
 - Section 5.6 – ICMP: The Internet Control Message Protocol



Wireshark lab

See Wireshark Lab#4 in the Home Assignment folder.

Due date: before next lecture
(Thursday Sep.24 13:00am)